

Supporting Information

Efficient promoters and reaction paths in the CO₂ hydrogenation to light olefins over zirconia supported iron catalysts

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Conversion and selectivity calculations:

The reaction feed and product analyses are based on gas chromatography (GC). The A indicates the peak area in the GC, n means moles of products (in terms of carbon atom moles), m represents the number of carbon atoms and x and y indicate the number of carbon atoms and the number of hydrogen atoms in any one hydrocarbon and S shows products selectivity.

The CO₂ conversion was calculated as follows:

$$X_{CO_2}(\%) = \left((F_{CO_2 \text{ in}} - F_{CO_2 \text{ out}}) / F_{CO_2 \text{ in}} \right) \cdot 100$$

where $F_{CO_2 \text{ in}}$ and $F_{CO_2 \text{ out}}$ are the molar flows of CO₂ at the inlet and the outlet of the reactor, respectively.

The selectivities were calculated on carbon basis. The selectivity to CO was calculated using the formula below:

$$S_{CO}(\%) = \left(F_{CO} / (F_{CO_2 \text{ in}} - F_{CO_2 \text{ out}}) \right) * 100$$

where “F_{CO}” refers to the molar flow of CO at the reactor outlet.

The selectivities to hydrocarbons (excluding CO) were calculated using the following expression:

$$S_i(\%) = \left(F_i / (F_{CO_2 \text{ in}} - F_{CO_2 \text{ out}} - F_{CO}) \right) * 100$$

where F_i refers to the carbon-based molar flow of hydrocarbons.

Table S1. Amount of desorbed CO₂ measured by TPD

Catalyst	CO ₂ amount (μmol/g)			
	Peak 1 (80-100°C)	Peak 2 (150-220°C)	Peak 3 (350°C)	Peak 4 (420°C)
ZrO₂	59.3	132.4	-	-
Fe/ZrO₂	15.4	26.4	-	-
FeK/ZrO₂	47.7	78.6	292.2	-
FeKCe/ZrO₂	29.6	42.4	188.4	-
FeKCs/ZrO₂	56.4	89.4	332.7	-
FeKCu/ZrO₂	26.5	60.2	127.6	207.9
FeKMo/ZrO₂	32.4	53.7	-	-
FeKGa/ZrO₂	42.9	47.8	236.1	-

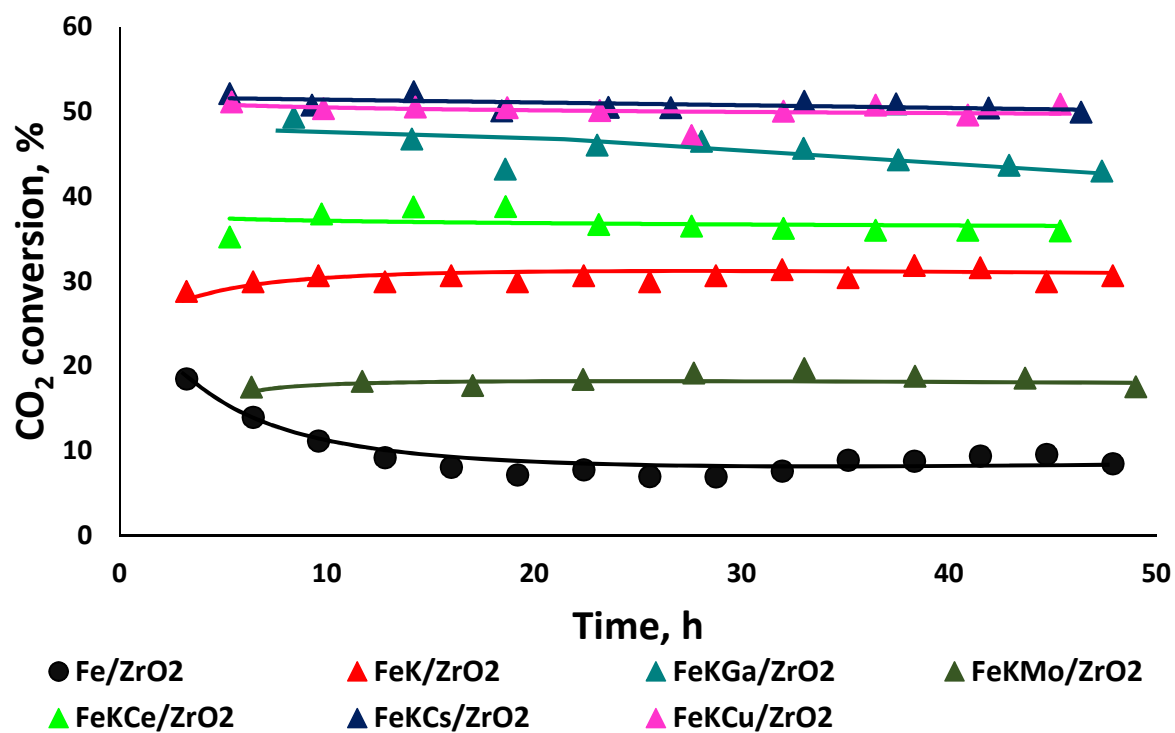


Figure S1. CO₂ conversion as a function of time for iron catalysts promoted with K, Ga, Mo, Ce, Cs, and Cu. Reaction conditions: T = 350 °C, P = 10 bar, H₂/CO₂ = 3, GHSV = 9.25 L g⁻¹ h⁻¹.

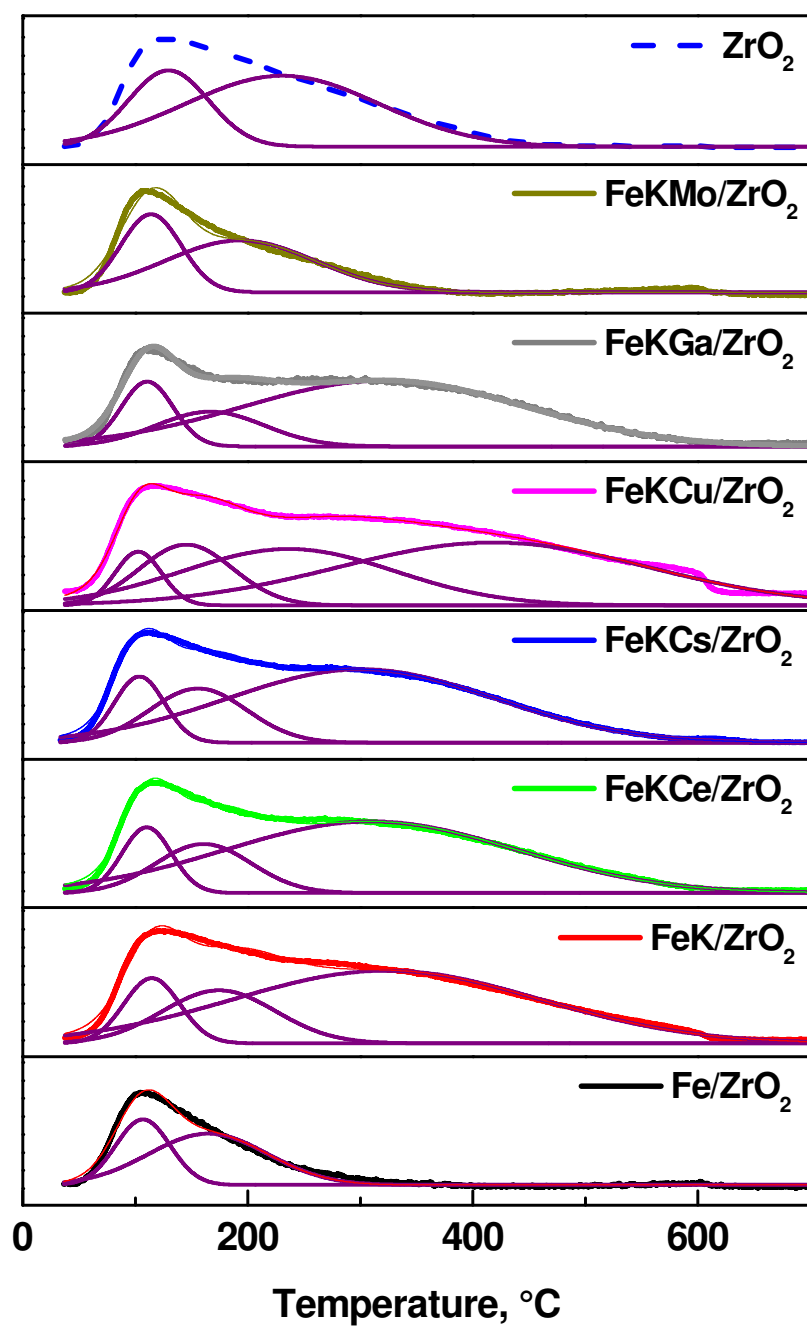
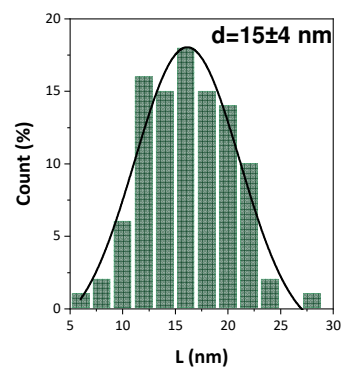
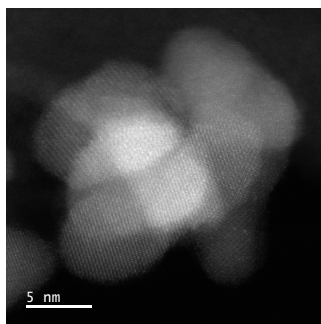
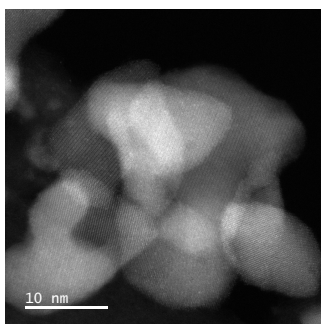
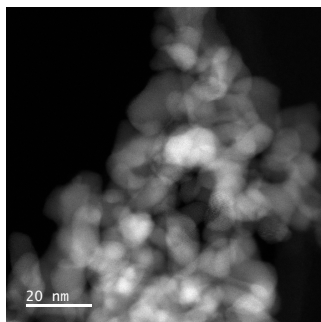
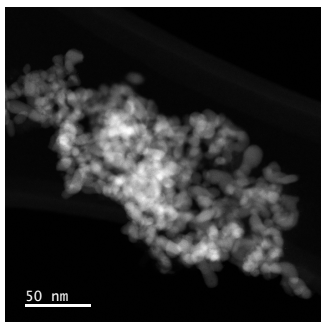
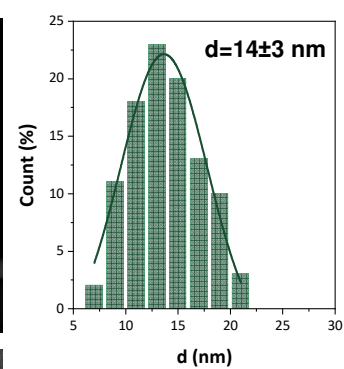
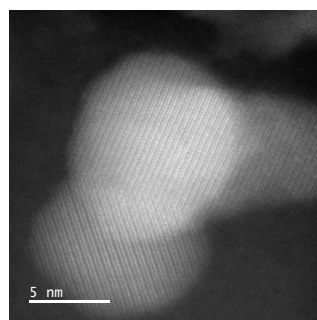
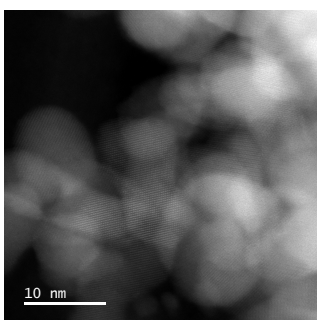
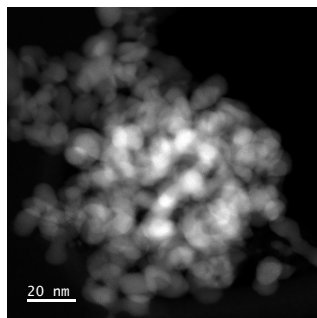
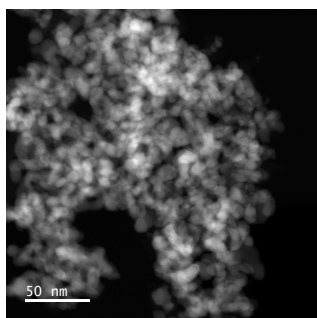


Figure S2. CO₂-TPD curves deconvolution for different ZrO₂-supported catalysts.

a



b



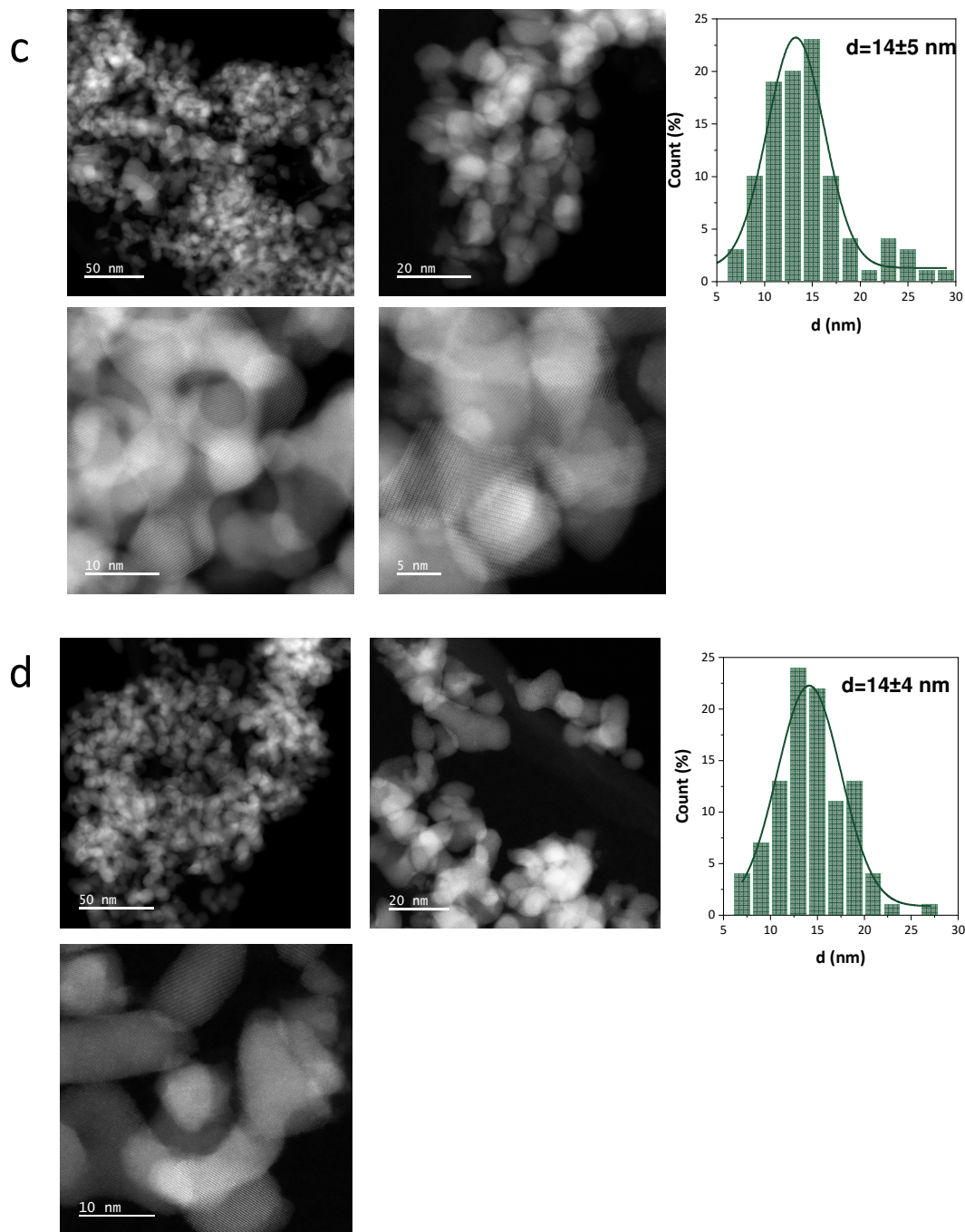
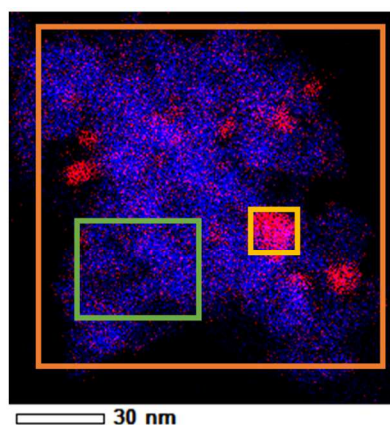
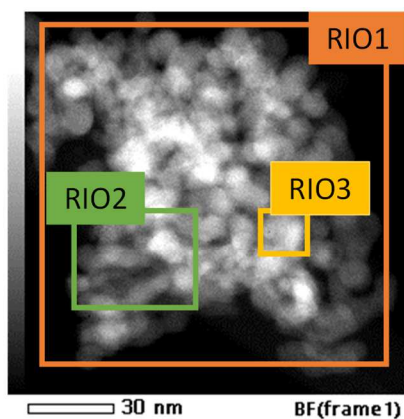


Figure S3. STEM-HAADF and histograms of catalyst nanoparticle sizes for the activated Fe/ZrO₂ (a), FeK/ZrO₂ (b), FeKGa/ZrO₂ (c) and FeKMo/ZrO₂ (d) catalysts

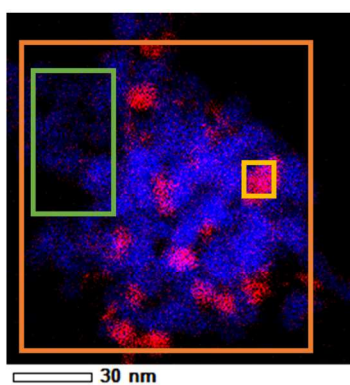
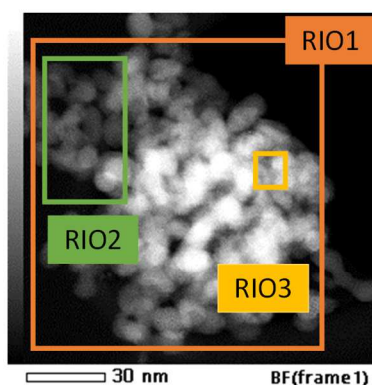
a



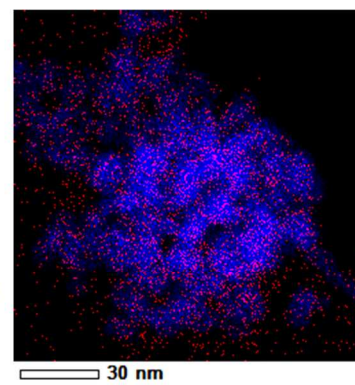
Superimposed
image : O, Fe

Element	RIO1		RIO2		RIO3	
	Weigh%	Atomic %	Weight%	Atomic %	Weight%	Atomic %
O	15.29	48.72	17.73	53.73	4.42	17.03
Fe	11.12	10.15	7.54	6.55	42.77	47.25
Zr	73.59	41.12	74.73	39.72	52.81	35.72

b

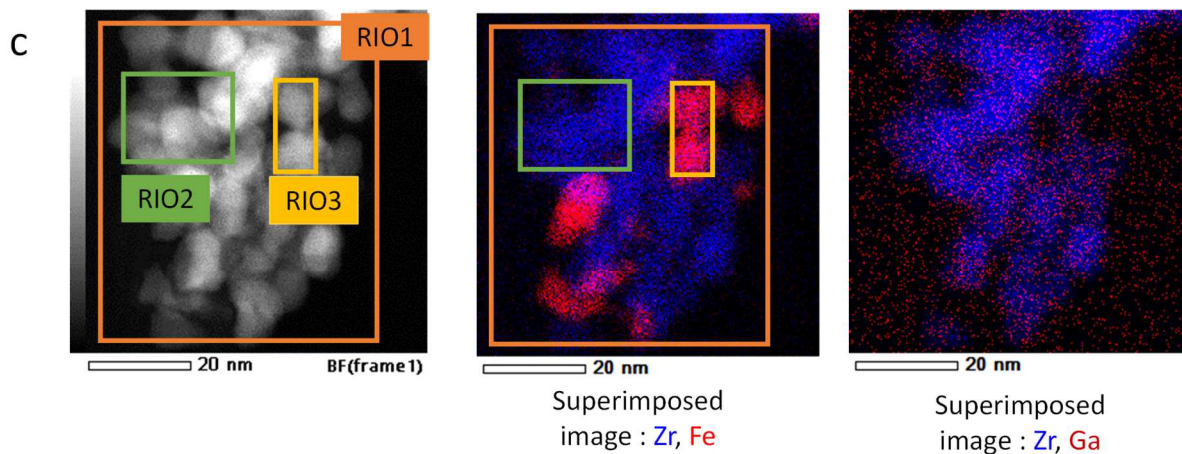


Superimposed
image : Zr, Fe

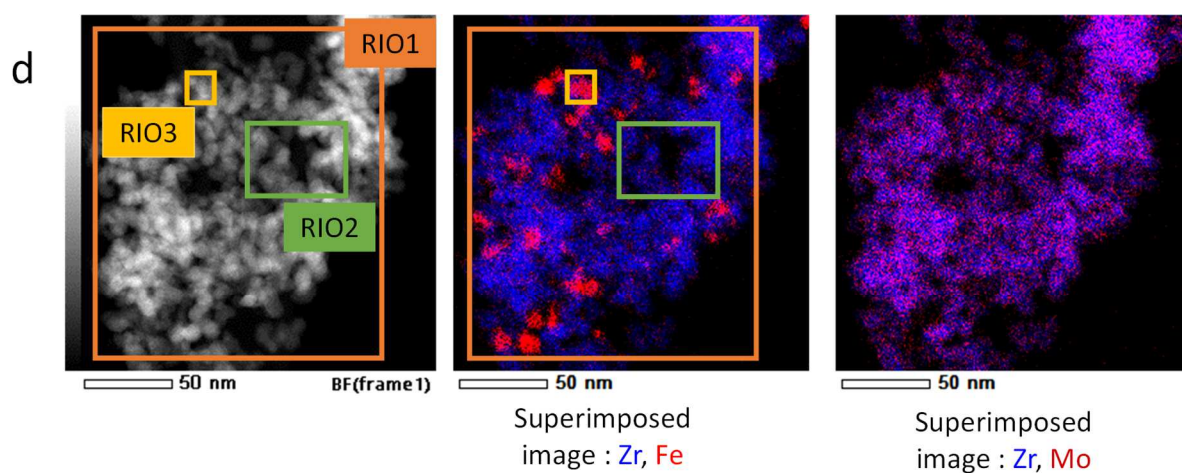


Superimposed
image : Zr, K

Element	RIO1		RIO2		RIO3	
	Weigh%	Atomic %	Weight%	Atomic %	Weight%	Atomic %
K	0.02	0.02	0.04	0.04	0	0
O	12.29	42.01	18.36	54.97	0.69	2.49
Fe	14.26	13.96	6.41	5.50	84.33	87.94
Zr	73.43	44.01	75.19	39.48	14.99	9.57



Element	RIO1		RIO2		RIO3	
	Weigh%	Atomic %	Weight%	Atomic %	Weight%	Atomic %
Zr	66.31	41.25	75.18	39.95	38.19	26.02
O	9.69	34.36	17.81	53.97	1.87	7.28
Fe	23.97	24.36	6.99	6.07	59.91	66.68
K	0.01	0.02	0.01	0.01	0.01	0.01
Ga	0.01	0.01	0.01	0.00	0.02	0.02



Element	RIO1		RIO2		RIO3	
	Weigh%	Atomic %	Weight%	Atomic %	Weight%	Atomic %
Zr	71.16	42.28	74.19	40.99	18.41	12.00
O	12.80	43.36	16.49	51.94	0.85	3.16
Fe	13.02	12.64	5.76	5.19	78.04	83.14
K	0.03	0.04	0.01	0.01	0.02	0.03
Mo	3.00	1.69	3.56	1.87	2.68	1.66

Figure S4. EDS analysis of iron-rich and iron-poor areas in the activated Fe/ZrO₂ (a), FeK/ZrO₂ (b), FeKGa/ZrO₂ (c) and FeKMn/ZrO₂ (d) catalysts

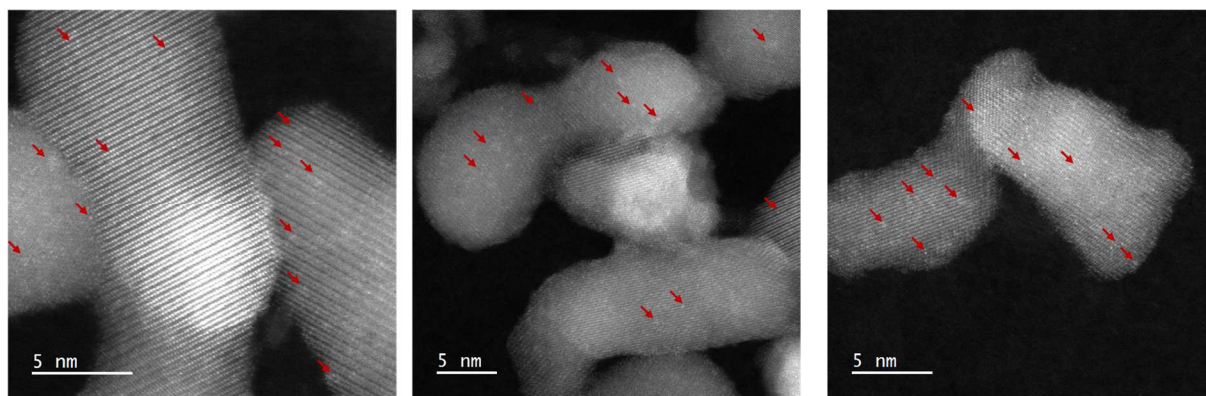


Figure S5. Atoms and atomic clusters of Mo in activated FeKMo/ZrO₂ visible as light spots and agglomerations shown by arrows (Z-contrast in STEM-HAADF mode)

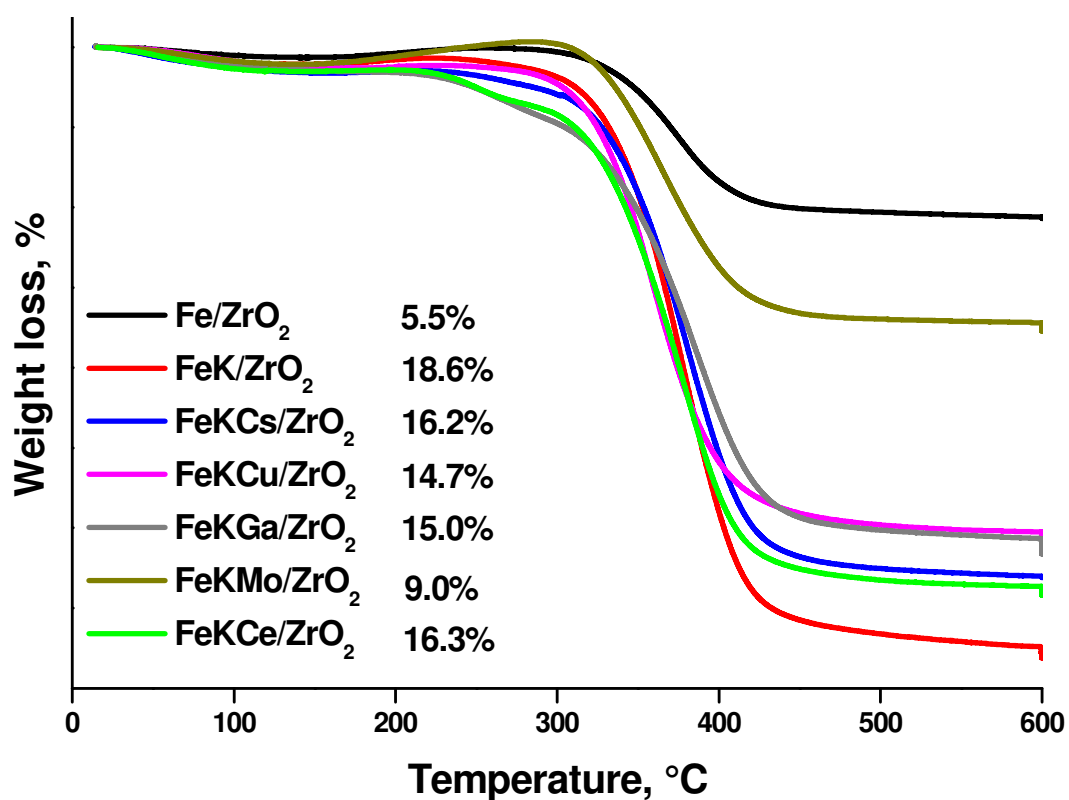


Figure S6. Thermogravimetric analysis (TGA) curves of the spent catalysts.